MILITARY SPECIFICATION

REGULATOR, PRESSURE, RIOT CONTROL AGENT DISPERSER

This specification is approved for use within U.S. Army Chemical Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of pressure regulator.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Standards. The following standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of US Military Property.
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines.

2.1.2 Other Government documents and drawings. The following other Government documents and drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research and Development Center, ATTN: SMCCR-SPD-TS, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1040

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DRAWINGS AND TECHNICAL DATA PACKAGE LIST (TDPL)

U.S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH AND DEVELOPMENT CENTER

TDPL 116-3-92 - Regulator, Pressure, Riot Control Agent Disperser.

(Copies of standards, drawings and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies of these freight classification ratings, rules, and regulations should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials used in the manufacture of the regulator shall conform to the specifications forming a part of this specification. Materials not definitely specified shall be at an accepted commercial type and grade suitable for the purpose intended. Material shall be free from defects and imperfections that might affect serviceability of the assembled product.

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3.1.2 Components. All components shall be of an accepted commercial type and of the manufacturer's latest design.

3.2 Design. The regulator shall be complete and capable of operation under the conditions specified in this specification. All regulators procured under this specification from one contractor shall be physically and mechanically identical (Drawing 116-3-92). Regulators shall be of a design not hazardous to personnel. The regulator shall be capable of maintaining its rated functioning for 50,000 cycles (a cycle is designated as a 3 second flow of regulated air). The regulator shall have an adjustment feature to obtain the setting specified in 3.3. The adjustment shall be suitably designed to discourage tampering. The regulator shall be capable of operating at temperatures of -40° to +155°F. The weight of the regulator shall not exceed 1.0 pound.

3.3 Functioning and leakage. The regulator shall not leak and shall maintain an outlet pressure of 180 ± 20 pounds per square inch gage (psig) at a flow rate of 0 to 30 standard cubic feet per minute (scfm) [(40°F at 14.7 pounds per square inch (psi)] of clean air or nitrogen at inlet pressures of 360 ± 50 psig and 2100 (+100 or -0) pisg and there shall be no pressure rise for the last minute of a three minute period when tested in accordance with 4.4.4.1.

3.4 High temperature. The regulator shall conform to 3.3 when tested at +155°F as specified in 4.3.3.1.

3.5 Low temperature. The regulator shall conform to 3.3 when tested at -40°F as specified in 4.3.3.1.

3.6 Temperature shock. The regulator shall conform to 3.3 when tested as specified in 4.3.3.1.

3.7 Salt fog. The regulator shall conform to 3.3 when tested as specified in 4.3.3.1.

3.8 Dust. The regulator shall conform to 3.3 when tested as specified in 4.3.3.1.

3.9 Proof pressure. The regulator shall not leak and shall be free from permanent deformation and damage when subjected to a minimum pressure of 3300 psig as specified in 4.4.4.2 (see 6.3).

3.10 Cyclic functioning. The regulator shall conform to 3.3 when cycled in accordance with table I and tested as specified in 4.3.3.2.

3.11 Vibration. The regulator shall conform to 3.3 after testing as specified in 4.3.3.3.

3.12 Mechanical shock. The regulator shall conform to 3.3 after testing as specified in 4.3.3.4.

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TABLE I. Criteria for failure rate testing

Column I Number of failures	Column II Cumulative time (cycles) (see 3.2)
1	30,000
2	48,000
3	63,000
4	78,000
5	92,000
6	105,000
7	119,000
8	132,000
9	145,000
10	157,000

The above cumulative operating times in colume II shall be met before the indicated number of failures in column I has occurred. (Cyclic functions shall be cumulative for all regulators cycled.) The regulator shall demonstrate a mean life of at least 10,000 cycles, at a 95 percent confidence level.

3.13 Identification. The regulator shall be identified by a nameplate in accordance with MIL-STD-130.

3.14 Preproduction. Prior to the start of regular production, a preproduction sample of regulators shall be produced in accordance with this specification for examination and tests.

3.15 Workmanship. The regulator shall be free from foreign matter (oil, dirt, chips and grease) and damage that could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program.

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The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Lot. A preproduction sample of 63 regulators shall be manufactured using the same methods, materials, equipment, and processes as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination and nondestructive tests. The preproduction lot shall be examined and tested in accordance with the classification of defects and with MIL-STD-105.

4.3.2.2 For destructive tests. The preproduction lot shall be tested in accordance with table II.

TABLE II. Acceptance/rejection criteria

Quantity	Requirement	Test	Accept (No. Defects)	Reject (No. Defects)
5	3.2	4.3.3.5	0	1
10	3.4	4.3.3.1	0	1
10	3.5	4.3.3.1	0	1
10	3.6	4.3.3.1	0	1
3	3.7	4.3.3.1	0	1
3	3.8	4.3.3.1	0	1
10	3.9	4.4.4.2	0	1
2 (min)	3.10	4.3.3.2	see table I	
5	3.11	4.3.3.3	0	1
5	3.12	4.3.3.4	0	1

4.3.2.3 Acceptance/rejection criteria. The contractor shall obtain written approval from the contracting activity prior to proceeding with regular production. No regulators subjected to a destructive-type test shall be included in any regular production quantity offered for acceptance.

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4.3.3 Tests.

4.3.3.1 High temperature, low temperature, temperature shock, salt fog, and dust. The regulator shall be tested in accordance with methods 501, procedure I, 502, 503, 509 and 510 respectively of MIL-STD-810.

4.3.3.2 Cyclic functioning. The regulator shall be mounted in a suitable fixture and shall operate to complete a minimum of 10,000 cycles.

4.3.3.3 Vibration. The regulator shall be mounted in a rigid fixture that simulates actual regulator installation. This assembly shall be securely fastened to the vibration table and vibrated in accordance with category (g), procedure x, method 514 of MIL-STD-810.

4.3.3.4 Mechanical shock. Testing except as follows shall be in accordance with Method 516, Procedure I of MIL-STD-810.

(a) Regulator. The regulator shall be mounted in a fixture that simulates actual regulator installation. The regulator shall be subjected to three shocks in each direction applied along three mutually perpendicular axes for a total of 18 shocks. The shock pulse shall be in accordance with figure 516-1 of MIL-STD-810 with a peak value (P) of 15 g and nominal duration (D) of 11 milliseconds.

(b) Repair parts. The packaged regulator (see 5.4) shall be mounted on the test plate and each of the six sides shall be subjected to three shocks in each direction along the three mutually perpendicular axes for a total of 18 shocks. The shock pulse shape shall be as above (4.3.3.4(a) with a peak value of 30 g with a normal duration of 11 milliseconds.

4.3.3.5 Weight. The weight of the regulator shall be verified by a weighing device accurate to 0.10 pound.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of regulators produced by one manufacturer, at one plant, from the same materials and under essentially the same manufacturing conditions.

4.4.2 Sampling for examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.3 Inspection procedure.

4.4.3.1 For examination. Sample regulators shall be examined in accordance with the classification of defects and with MIL-STD-105.

4.4.3.2 Critical defects. Each regulator in the lot shall be tested for the critical characteristics listed in the classification of defects.

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4.4.3.3 Classification of defects.

(a) Regulator (Drawing 116-3-92).

Categories	Defects	Acceptance standards
Critical:		
1	Functioning and leakage	4.4.4.1
2	Proof pressure	4.4.4.2
Major:	AQL 1.0 percent defective	
101	Diameter and location of installation holes incorrect	
102	Hole for inlet incorrect or incorrectly located	
103	Hole for outlet incorrect or incorrectly located	
104	Workmanship (3.15)	

4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Functioning and leakage.

(a) With 2200 psig air or nitrogen applied to the inlet and approximately 15 cubic inches downstream, flow 30 scfm to atmosphere and monitor downstream pressure.

(b) Reduce flow to zero and monitor for downstream pressure rise the last minute of a three minute period.

(c) Monitor for external leakage during (b) above using a suitable leak detecting solution.

(d) Repeat (a) and (b) above with an inlet pressure of 360 psig (see 6.3).

4.4.4.2 Proof pressure. With regulator ports manifolded, but separated by a hand valve, apply 270 psig air or nitrogen to the manifold. Shut separation valve and continue pressurizing the inlet port to 3300 psig. Hold for two minutes. Depressurize only to 270 psig, open the separation valve and bleed to zero (g) psig (see 6.3).

5. PACKAGING

5.1 Packaging (interplant shipment). The regulators shall be packaged, 1 per container, in containers which will protect the item from damage in delivery from the supply source to a first receiving activity.

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5.2 Packing (interplant shipment). The regulators, packaged in accordance with 5.1, shall be packed to be in compliance with Uniform Freight Classification and other rules and regulations applicable to the mode of transportation and to provide adequate protection from damage in delivery from the supply source to a first receiving activity.

5.3 Marking. In conjunction with information furnished in the contract or order, packaging and packing containers shall be marked in accordance with the applicable requirements of MIL-STD-129.

5.4 Repair parts. When the pressure regulator is being procured for stockage and issue as a repair part, preservation, packaging, packing and marking shall be as specified on the repair parts packaging data sheet (see 6.4).

6. NOTES

6.1 Intended use. This item is intended for use as a component of the M33 dispersers.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preproduction:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether to proceed with production.

6.3 Caution. Contractors are advised to take necessary precautions to protect personnel from injury during test operations.

6.4 Packaging data sheet. The packaging data sheet is identified by the Federal Stock Number for the item and should be ordered from Commander, U.S. Army Chemical Research and Development Center, ATTN: SMCCR-SPD-PP, Aberdeen Proving Ground, MD 21010-5423.

6.5 International interest. Certain provisions of this specification are the subject of international standardization agreement ABCA-Army Quadripartite Standardization Agreement - 317. When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accomodations.

6.6 Supersession data. This specification supersedes and includes the requirements of Edgewood Arsenal Purchase Description EA-R-1076, dated 11 September 1970 and Amendment 1 to EA-R-1076, dated 26 August 1971.

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Custodian:

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Preparing activity:

Army - EA

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-51516(EA)	REGULATOR, PRESSURE, RIOT CONTROL AGENT DISPERSER

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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